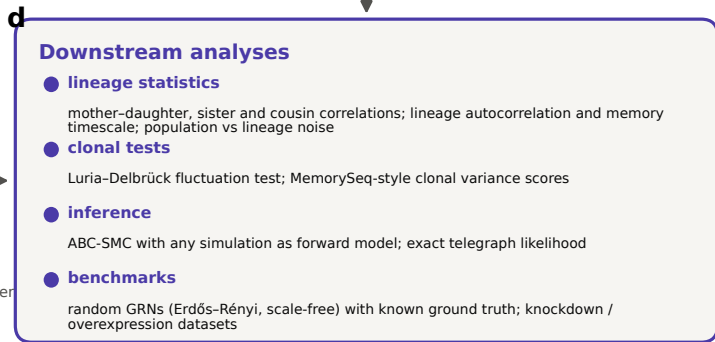
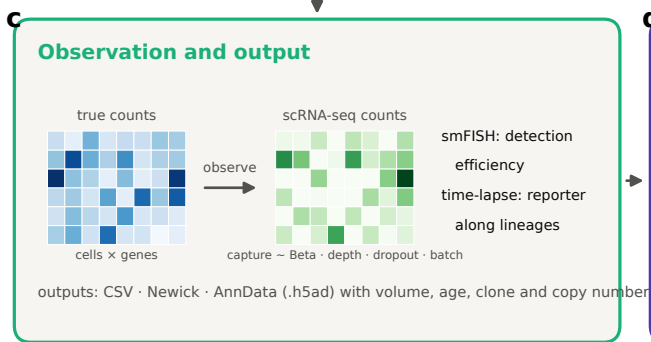
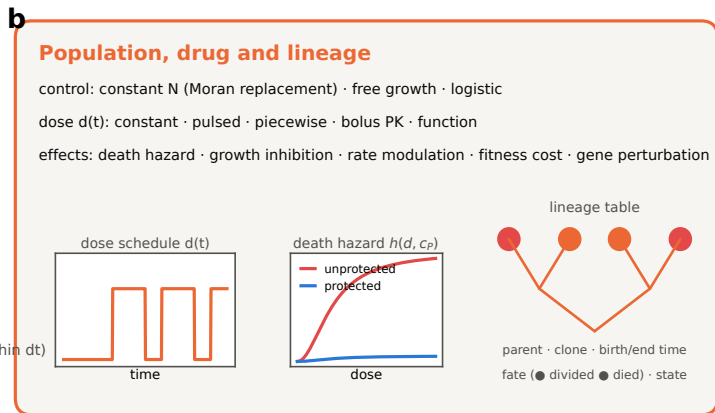
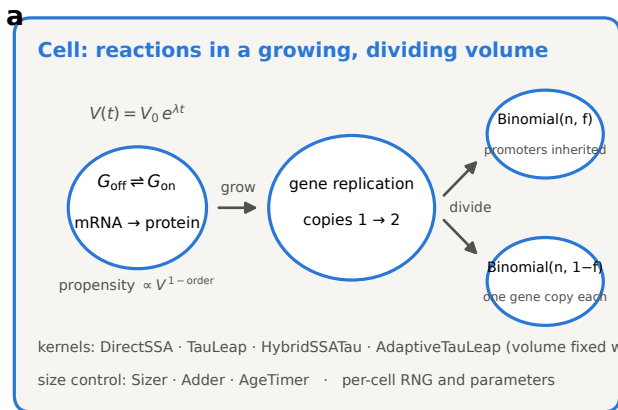




e A complete example: a resistance gene in a growing, dividing, drug-treated population

```
using Biomodeling, Random
model = telegraph_model(k_on = 0.002, k_off = 0.02, k_tx = 30.0, k_dm = 1.0, k_tl = 4.0, k_dp = 0.2) # resistance gene R → mRNA → P
x0 = initial_state(model; G_off = 1)
pert = Perturbation(PiecewiseDose([0.0, 40.0], [0.0, 1.0])); # drug from t = 40
effects = [DeathHazard(h_max = 0.15, EC50 = 0.5, m = 2.0, protect = :protein, K = 150.0, q = 4.0)]
st = PopulationSettings(dt = 0.1, growth = ExponentialGrowth(log(2)/20), size_control = Sizer(2.0; cv = 0.05),
    replication = Replication(0.5), control = FreeGrowth(max_cells = 20_000))
res = simulate_population(model, x0, 500, (0.0, 200.0); settings = st, perturbation = pert, rng = Xoshiro(1))
heritability(res, :protein; relation = :sisters) # expression memory across divisions
Y = sequence(sample_cells(res; n = 500).counts, SeqProtocol(capture = 0.15)).Y # synthetic scRNA-seq
```